SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 30%

Total Marks: 100 **Time: 90 Mins**

No. of Questions: 1

Annexure A

Research Review & Analysis

Code of Conduct:

I _____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	**Decision Tree Learning:** Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	**Neural Networks vs Statistical Learning:** Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	**Kernel Methods:** Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	**Reinforcement Learning:** Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	**Explainable and Generalizable AI:** Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning

Structure of the Report:

S.No.	Paper Section	Expected Content
1	**Title**	Specific and concise title related to the selected research question/topic.
2	**Abstract**	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	**Keywords**	4–6 important keywords related to the selected topic.
4	**1. Introduction**	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	**2. Research Problem and Objectives**	Clearly state the research problem and specific objectives of the review.
6	**3. Literature Review**	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	**4. Comparative Analysis**	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	**5. Critical Analysis**	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	**6. Research Gap**	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	**7. Proposed Research Direction**	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	**8. Discussion**	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	**9. Conclusion**	Summarize the major findings, key insights, research gap, and future direction.
13	**References**	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.

Question 1: Decision Tree Learning

Title

A Comparative Review and Analysis of ID3, C4.5, and CART Decision Tree Learning Algorithms

Abstract

Decision Tree Learning is an important supervised machine learning technique used for classification and regression problems. Among the well-known decision tree algorithms, **ID3, C4.5, and CART** use different strategies for selecting attributes and constructing trees. This review analyzes these algorithms based on attribute-selection methods, pruning strategies, interpretability, computational complexity, handling of data types, and classification performance. ID3 uses **Information Gain**, C4.5 uses **Gain Ratio**, while CART commonly uses the **Gini Index** for classification. Research indicates that ID3 is simple and easy to understand but has limitations with continuous attributes and overfitting. C4.5 improves upon ID3 by supporting continuous attributes and pruning. CART generates binary trees and provides a flexible framework for both classification and regression. Recent comparative research also shows that the choice of splitting criterion can influence accuracy, tree depth, number of leaves, and construction time.

The major research gap is the need for decision-tree methods that simultaneously provide **high predictive performance, low computational cost, strong generalization, and clear interpretability** for modern and large-scale datasets.

Keywords: Decision Tree, ID3, C4.5, CART, Information Gain, Gain Ratio

1. Introduction

Decision Tree Learning is a supervised machine learning approach that represents decisions in the form of a tree structure. A decision tree consists of a **root node, internal decision nodes, branches, and leaf nodes**. Each internal node represents a decision based on an attribute, while a leaf node represents the final prediction.

Decision trees are popular because their decision-making process can be represented using simple rules such as **IF–THEN** conditions. This makes them relatively easy for users to understand compared with many complex machine learning models.

Three important decision-tree algorithms are:

1. **ID3 – Iterative Dichotomiser 3**
2. **C4.5**
3. **CART – Classification and Regression Trees**

ID3 was developed by Ross Quinlan and uses Information Gain for selecting attributes. C4.5 is an extension of ID3 that introduces improvements such as Gain Ratio and support for continuous attributes. CART uses binary splitting and commonly uses the Gini Index for classification.

2. Research Problem and Objectives

2.1 Research Problem

Different decision-tree algorithms use different attribute-selection and tree-construction strategies. Therefore, their accuracy, computational requirements, interpretability, scalability, and ability to generalize to unseen data can differ.

The research problem is:

How do ID3, C4.5, and CART differ in attribute selection, pruning, computational complexity, interpretability, and classification performance, and which approach is more suitable for different machine learning applications?

2.2 Objectives

The main objectives are:

- To study the working principles of ID3, C4.5, and CART.
- To compare their attribute-selection techniques.
- To analyze their pruning strategies.
- To compare computational complexity and scalability.
- To evaluate their interpretability.
- To analyze their classification performance.
- To identify limitations and research gaps.
- To propose possible improvements for future decision-tree systems.

3. Literature Review

3.1 ID3 Algorithm

ID3 is one of the classical decision-tree learning algorithms. It uses **Entropy** and **Information Gain** to determine the attribute that should be selected for splitting.

The entropy of dataset Scan be represented as:

$$Entropy(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

Information Gain is:

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

The attribute having the highest Information Gain is selected as the splitting attribute.

Advantages

- Simple to understand.
- Easy to implement.
- Produces interpretable decision rules.
- Suitable for categorical data.

Limitations

- Original ID3 does not naturally handle continuous attributes.
- It is susceptible to overfitting.
- Information Gain can favor attributes with many distinct values.

- Classical ID3 does not include a dedicated pruning mechanism.

3.2 C4.5 Algorithm

C4.5 was developed as an improvement over ID3. Instead of directly using Information Gain, it uses **Gain Ratio** to reduce the bias toward attributes having many possible values.

$$GainRatio(A) = \frac{Gain(A)}{SplitInfo(A)}$$

Advantages

- Handles categorical and continuous attributes.
- Uses Gain Ratio.
- Provides pruning.
- Can deal with missing values in its original formulation.
- Produces understandable decision rules.

Limitations

- More computationally demanding than the basic ID3 approach.
- Large datasets can increase tree-construction cost.
- Performance can still depend on parameter and dataset characteristics.

3.3 CART Algorithm

CART stands for **Classification and Regression Trees**. Unlike ID3 and the traditional C4.5 tree structure, CART constructs **binary decision trees**, meaning each internal node generally produces two branches.

For classification, CART commonly uses the **Gini Index**:

$$Gini(S) = 1 - \sum_{i=1}^{c} p_i^2$$

A lower Gini impurity indicates a purer split.

CART can be used for both:

- Classification
- Regression

CART also incorporates pruning using a complexity-based approach.

Advantages

- Supports classification and regression.
- Binary structure is relatively systematic.
- Uses Gini impurity for classification.
- Supports pruning.
- Widely applicable to different datasets.

Limitations

- Binary splitting may create deeper trees.
- Can overfit without suitable regularization or pruning.
- The selected split can be affected by characteristics of the dataset.

4. Comparative Analysis

The uploaded assessment specifically expects comparison using parameters such as **accuracy, complexity, interpretability, computational cost, scalability, and generalization**.

Parameter	**ID3**	**C4.5**	**CART**
Attribute Selection	Information Gain	Gain Ratio	Gini Index
Tree Structure	Multiway	Multiway / rule-based representation	Binary
Continuous Attributes	Limited in original form	Supported	Supported
Pruning	No classical built-in pruning	Yes	Yes
Classification	Yes	Yes	Yes
Regression	No	Mainly classification	Yes
Interpretability	High	High	High
Overfitting	Relatively high risk	Reduced through pruning	Reduced through pruning
Computational Cost	Relatively low	Higher than ID3	Moderate
Scalability	Suitable for smaller/moderate datasets	Suitable but can become computationally expensive	Good flexibility
Main Strength	Simplicity	Improved flexibility	Classification + regression

5. Critical Analysis

ID3 – Critical Analysis

Strengths:

- Simple algorithm.
- Easy to interpret.
- Easy to implement.
- Information Gain provides an intuitive attribute-selection mechanism.

Weaknesses:

- Sensitive to attributes with many values.
- Original method is mainly designed for categorical attributes.
- Can generate unnecessarily large trees.
- Overfitting can occur.

C4.5 – Critical Analysis

Strengths:

- Improves upon ID3.
- Gain Ratio reduces the multiple-value attribute bias.
- Handles continuous attributes.
- Pruning improves generalization.
- Produces understandable rules.

Weaknesses:

- More computationally expensive than the basic ID3 approach.

- Large datasets may increase processing requirements.
- The final tree can still become complex depending on the dataset.

CART – Critical Analysis

Strengths:

- Supports both classification and regression.
- Uses binary splits.
- Uses Gini impurity for classification.
- Provides pruning.
- Flexible for different data-analysis tasks.

Weaknesses:

- Can generate deep trees.
- Requires suitable pruning or regularization to control overfitting.
- Its performance depends strongly on the quality and distribution of the input data.

A recent comparative study notes that CART is versatile but can overfit without careful parameter tuning and pruning, while ID3 is simple but more limited in handling numerical and missing data; C4.5 offers broader attribute support and pruning.

6. Research Gap

Based on the reviewed research, the following research gaps can be identified:

Gap 1 – Overfitting

Decision trees can produce complex structures when trained on noisy or highly detailed datasets.

Gap 2 – Splitting-Criterion Bias

Different splitting criteria can favor different types of attributes. Therefore, one criterion may not be optimal for every dataset.

Gap 3 – Large-Scale Data

As dataset size increases, tree construction and parameter optimization can become computationally expensive.

Gap 4 – Accuracy vs Interpretability

More complex models may improve predictive performance but can reduce the simplicity that makes decision trees attractive.

Gap 5 – Modern Data Characteristics

Modern datasets may contain high-dimensional, continuous, noisy, imbalanced, and missing data. A single classical decision-tree algorithm may not perform optimally under all these conditions.

A 2023 comparative study evaluated six splitting approaches across 12 UCI datasets using accuracy, tree depth, leaf-node count, and development time, demonstrating that splitting criteria can have a substantial effect on decision-tree behavior.

7. Proposed Research Direction

A possible improvement is to develop an **Adaptive Hybrid Decision Tree Algorithm**.

Proposed approach

Dataset
↓
Data Preprocessing
↓
Feature Quality Analysis
↓
Adaptive Selection of Splitting Criterion
↓
Decision Tree Construction
↓
Pruning
↓
Cross-Validation
↓
Performance Evaluation

The proposed system could dynamically select an appropriate splitting criterion based on dataset characteristics rather than always using a single fixed criterion.

For example:

- Information Gain → suitable for simple categorical problems.
- Gain Ratio → useful when attributes have many possible values.
- Gini Index → useful for classification-based binary splitting.

The final model can then be evaluated using accuracy, precision, recall, F1-score, tree depth, training time, and inference time.

8. Discussion

The literature shows that **ID3, C4.5, and CART are based on the same general decision-tree concept but differ significantly in their splitting and pruning mechanisms**.

ID3 provides the simplest approach, but its limitations make it less suitable for complex modern datasets. C4.5 addresses several ID3 limitations through Gain Ratio, continuous-attribute handling, and pruning. CART provides a more flexible framework because it supports both classification and regression and constructs binary trees.

Therefore, there is no single algorithm that is universally best. The appropriate method depends on the dataset, computational resources, required interpretability, and application requirements.

9. Conclusion

ID3, C4.5, and CART are important decision-tree learning algorithms with different characteristics. **ID3 uses Information Gain, C4.5 uses Gain Ratio, and CART commonly uses Gini Index for classification.** C4.5 improves several limitations of ID3, while CART provides additional flexibility through classification and regression capabilities.

From the comparison, **C4.5 provides a strong balance between interpretability, handling of different attribute types, and generalization**, while CART is particularly useful when both classification and regression are required. ID3 remains valuable as a simple and educational decision-tree algorithm.

Future research should focus on adaptive splitting criteria, efficient pruning, improved handling of noisy and high-dimensional data, and methods that maintain high accuracy while preserving decision-tree interpretability.

References

1. Quinlan, J. R. (1986). **Induction of Decision Trees**. *Machine Learning*, 1, 81–106.
2. Quinlan, J. R. (1993). **C4.5: Programs for Machine Learning**. Morgan Kaufmann.
3. Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). **Classification and Regression Trees**. Wadsworth.
4. Aaboub, F., Chamlal, H., & Ouaderhman, T. (2023). **Statistical analysis of various splitting criteria for decision trees**. *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*.
5. **Influence of Explanatory Variable Distributions on the Behavior of the Impurity Measures Used in Classification Tree Learning**. Recent comparative research on CART, ID3, and C4.5 splitting criteria and their advantages and limitations.
6. **Splitting Choice and Computational Complexity Analysis of Decision Trees**. Research analyzing splitting criteria, computational complexity, ID3, C4.5, and CART.